

31/3/2025

28. calendar

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days $\rightarrow$ M, T, W, Th, F, St, Sn

month $\rightarrow$ Jan, _____, dec

year $\rightarrow$ Random / Leap Year

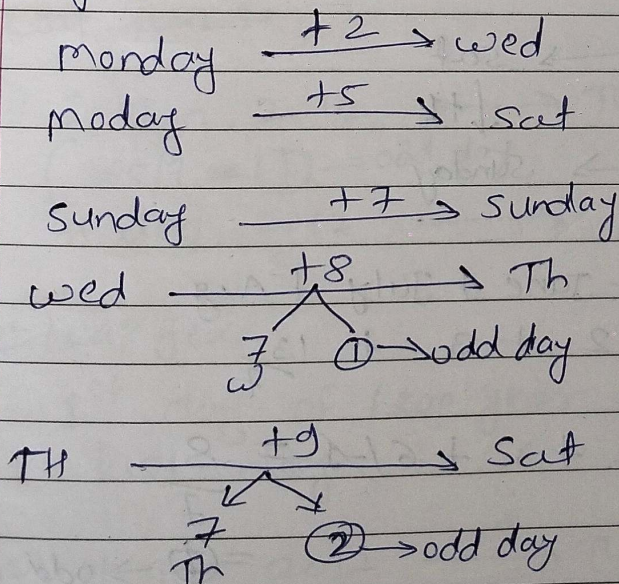
Date $\Rightarrow$ d/m/y

	M	T	W	TH	F	St	Sn
+7	1	2	3	4	5	6	7
+7	8	9	10	11	12	13	14
+7	15	16	17	18	19	20	21
+7	22	23	24	25	26	27	28
+7	29	30	31				

① month $\Rightarrow$ 30/31

② Each days Repeat after every 7 days.

④ days:-



Sat $\xrightarrow{+14}$ Sat

Sat $\xrightarrow{+16}$ Monday

①4 $\rightarrow$ ② $\rightarrow$ odd day

1 June $\xrightarrow{+29/7 \Rightarrow ①}$ odd day $\xrightarrow{+1}$ Tuesday

30 June $\xrightarrow{+1}$ wed

2 July $\xrightarrow{+25/7 \Rightarrow 4}$ Monday

27 July $\xrightarrow{+4}$ Friday

$$\frac{16+2+3}{21} + \frac{2+3}{19} + \frac{3+4}{17} = 51-2$$

(#) Year
 1 year $\xrightarrow{\text{ordinary year}} \frac{365}{7} \Rightarrow 52 \text{ week} + 1 \text{ d} \rightarrow \text{①}$ odd day
↓

1 Leap year $\rightarrow \frac{366}{7} \Rightarrow 52 \text{ week} + 2 \text{ d} \rightarrow \text{②}$ ↑
odd day

Ex:-

① 1 Jan 2001 $\rightarrow$ sun
 $\frac{365}{7} = \text{①}$
 31 dec 2001 $\rightarrow$ sun

Imp (#) ordinary Year

1st day is same as last day

② 1 Jan 2001 $\rightarrow$ sun
 $\left(\frac{365}{7} = \text{①} \right) + 1 \rightarrow \text{odd day}$
 1 Jan 2002 $\rightarrow$ mon

Leap years:-

① 1 Jan 2012 $\rightarrow$ Tuesday
 $\left(\frac{365}{7} = \text{①} \right) \rightarrow \text{odd day} \quad \downarrow +1$
 31 dec 2012 $\rightarrow$ wed

Imp # Leap year

1st day of Leap year is one extra day in last day

② 1 Jan 2012 $\rightarrow$ Tue
 $\left(\frac{366}{7} = \text{②} \right) \rightarrow \text{odd day} \quad \downarrow +2$
 1 Jan 2013 $\rightarrow$ Thu

1/4/2025

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① 5 April 2002 → sat

5^{Nov} dec 2010
27 dec 2011 → wed

+4

dec

years

$$22 + (2+3+2+3+3+2+3+2) + 1+1+1+2+1+1+1+2+1+1 = 4$$

Cleander Repeat → 400y 400y 400y

Imp

$$100y \rightarrow \frac{100+24}{7} - \frac{124}{7} \Rightarrow 5 \rightarrow \text{odd day}$$

$$200y \rightarrow 10 \rightarrow 3 \rightarrow \text{odd day}$$

$$300y \rightarrow 15 \rightarrow 11 \rightarrow \text{odd day}$$

$$400y \rightarrow 1+5+1 \Rightarrow 7 \rightarrow 0 \rightarrow \text{odd day}$$

400y 800y 1200y 1600y 2000y

monday → 1 Jan 2001

monday - 1
Tuesday - 2
Wednesday - 3
Thursday - 4
Friday - 5
Saturday - 6
Sunday → 7/0

(#) last day century
100y → 5 → Friday
200y → 3 → Wednesday
300y → 1 → Monday
400y → 0 → Sunday

① 15 Aug 1947 → Friday

$$\frac{46}{4} = 11$$

years $1600 \cdot 300 \cdot 46$

$$[0 + 1 + (46 + 11)] + 3 + 0 + 3 + 2 + 3 + 2 + 3 + 15$$

Jan Feb Mar April May June July Aug

$$\Rightarrow 12 \Rightarrow \textcircled{5} \rightarrow \text{odd day}$$

Friday → 5

② 26 Jan 1950 → Thursday

1600 300 49

$$[0 + 1 + (49 + 12)] + 26 \Rightarrow \frac{39}{7} \Rightarrow 4$$

Years → Leap years

4 → Thursday

③ 15 March ^{Leap} 2012 → Thursday
2011

2000 11 → Leap year

$$0 + (11 + 2) + 3 + 1 + 151 \rightarrow 4$$

year → Thursday

1) It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2010.
a) Sunday b) Saturday c) Friday d) Wednesday.

1 Jan 2006 → Sun
↓ +5

1 Jan ²⁰⁰⁹ 2010 → Friday

2006 2007 2008 2009

$$1 + 1 + 2 + 1 = \textcircled{5}$$

2) what was the day of the week on 28th May, 2006?

a) Thursday b) Friday c) Saturday d) Sunday

28 May 2006

(2005)

2000 5 Jan Feb March April May

$$0 + (5+1) + 3 + 0 + 3 + 2 + 28 \Rightarrow 0$$

Leap year

0 $\rightarrow$ Sunday

3) What was the day of the week on 17th June, 1998?

a) Monday b) Tuesday c) Wednesday d) Thursday

17 June 1998

(1997)

29
97
4

$$1600 + 300 + 97$$

$$[0 + 1 + (97+24)] + 3 + 0 + 3 + 2 + 3 + 17$$

$$\Rightarrow \frac{24}{7}$$

(3) — Wednesday

$$\Rightarrow 3$$

4) What will be the day of week 15th Aug 2010?

a) Sunday b) Monday c) Tuesday d) Friday

15 Aug 2010

(2009)

0 $\rightarrow$ Sunday

2000 9

Jan Feb March April May Jun July Aug

$$[0 + (9+2)] + 3 + 0 + 3 + 2 + 3 + 2 + 3 + 15$$

Leap years

$$\Rightarrow 0$$

- 5) Today is Monday. After 61 days, it will be;
 a) Wednesday ☒ b) Saturday c) Tuesday d) Thursday.

m $\xrightarrow{61/7 \rightarrow 5}$ (Sat)

- 6) If 6th March 2005 is Monday, what was the day of the week on 6th March 2004?
☒ a) Sunday b) Saturday c) Tuesday d) Wednesday

6 March 2004 $\rightarrow$ Sunday
 $\frac{365}{7} \rightarrow \text{①} \rightarrow \text{odd day}$
 6 March 2005 $\rightarrow$ Monday $\leftarrow \begin{matrix} -1 \\ +1 \end{matrix}$

- 7) On what dates of April, 2001 did Wednesday fall?

a) 1st, 8th, 15th, 22nd, 29th

☒ b) 2nd, 9th, 16th, 23rd, 30th

c) 3rd, 10th, 17th, 24th

d) 4th, 11th, 18th, 25th

2000 Jan Feb Mar April

0 + 3 + 2 + 3 + 1 $\Rightarrow$ 2 $\rightarrow$ Tuesday

wed $\rightarrow$ 2, 9, 16, 23, 30

$\nwarrow$
1 April 2001

- 8) How many days are there in x weeks x days?
 a) $7x^2$ ☒ b) $8x$ c) $14x$ d) 7

$$[x \times 7 + x] \Rightarrow 7x + x = \text{⑧}x$$

- 9) The last day of a century cannot be
 a) Monday b) Wednesday ☒ c) Tuesday d) Friday

2, 4, 6 → not last day of century
 Tuesday Thursday Saturday

- 10) On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?
 a) Tuesday b) Monday ☒ c) Sunday d) Wednesday

8 Feb 2004 → Sunday
 $\frac{366}{7} \rightarrow \text{odd day} \rightarrow 2$
 $\frac{366}{7} \rightarrow 8 \text{ Feb } 2005 \rightarrow \text{Tuesday}$
 Sunday $\xrightarrow{-2} \text{Tuesday} \xrightarrow{+2}$

- 11) The calendar for the year 2007 will be the same for the year:

a) 2014 b) 2016 ☒ c) 2017 d) 2018

2007 → 1 + 2 + 1 + 1 + 1 + 2 + 1 + 1 + 1 + 2 + 1 = 14
 14 ÷ 7 = 2
 2017

- 12) Which of the following is not a leap year?

☒ a) 700 b) 800 c) 1200 d) 2000

century divide by 400 → Leap year

- 13) On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec 2006?

a) Sunday b) Thursday c) Tuesday ☒ d) Friday

8 Dec 2007 → Sat
 $\frac{365}{7} \rightarrow 1 \text{ odd day} \rightarrow -1$
 8 Dec 2006 → Fri

ordinary year $\rightarrow +1$
Leap year $\rightarrow +2$

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14) January 1, 2008 is Tuesday. what day of the week lies on Jan 1, 2009?

a) Monday b) Wednesday c) Thursday d) Sunday

1 Jan 2008 $\xrightarrow{\text{Leap year}}$ Tuesday
 $\frac{366}{7} = 52 \text{ R } 2 \rightarrow \text{odd day} \quad \downarrow +2$
1 Jan 2009 $\rightarrow$ Thursday

15) January 1, 2007 was Monday. what day of the week lies on Jan, 1 2008?

a) Monday b) Tuesday c) Wednesday
d) Sunday

1 Jan 2007 $\rightarrow$ Monday
 $\frac{365}{7} = 52 \text{ R } 1 \rightarrow \text{odd day} \quad \downarrow +1$
1 Jan 2008 $\rightarrow$ Tuesday